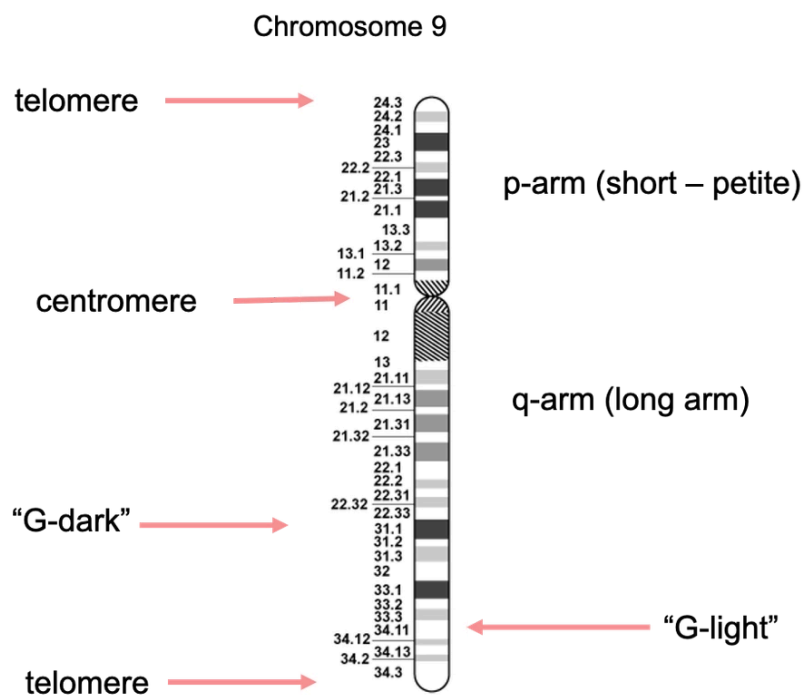


5.2 Chromosomal Abnormalities

- ▼ What is a karyotype?
 - An individual's **complete set of chromosomes**.
- ▼ What is the name of the stain used to view chromosomes?
 - **Gisema** stain
- ▼ Image showing structure of a chromosome.



- ▼ What does +/- mean in relation to mutations in genetics?
 - +/- **before** the chromosome number (i.e +8) means there has been a gain/loss of the **entire chromosome**
 - +/- **after** the chromosome number (i.e 4-) means there has been a gain/loss of a **part of the chromosome**

▼ Standard nomenclature in genetics.

abbreviation	Meaning
p	Short arm
q	Long arm
pter	Tip of short arm
qter	Tip of long arm
cen	centromer
del	deletion
der	Derivative chromosome (contains extra material)
dup	duplication
ins	insertion
inv	inversion
t	translocation
+/- Before chromosome number	Gain/loss whole chromosome
+/- After chromosome number	Gain/loss part of chromosome

▼ What is aneuploidy?

- An **abnormal number of chromosomes**

▼ Recall three important roles of meiosis.

- Random assortment
- Recombination
- Genetic variation
- The first two lead to the latter

▼ What happens in meiosis I?

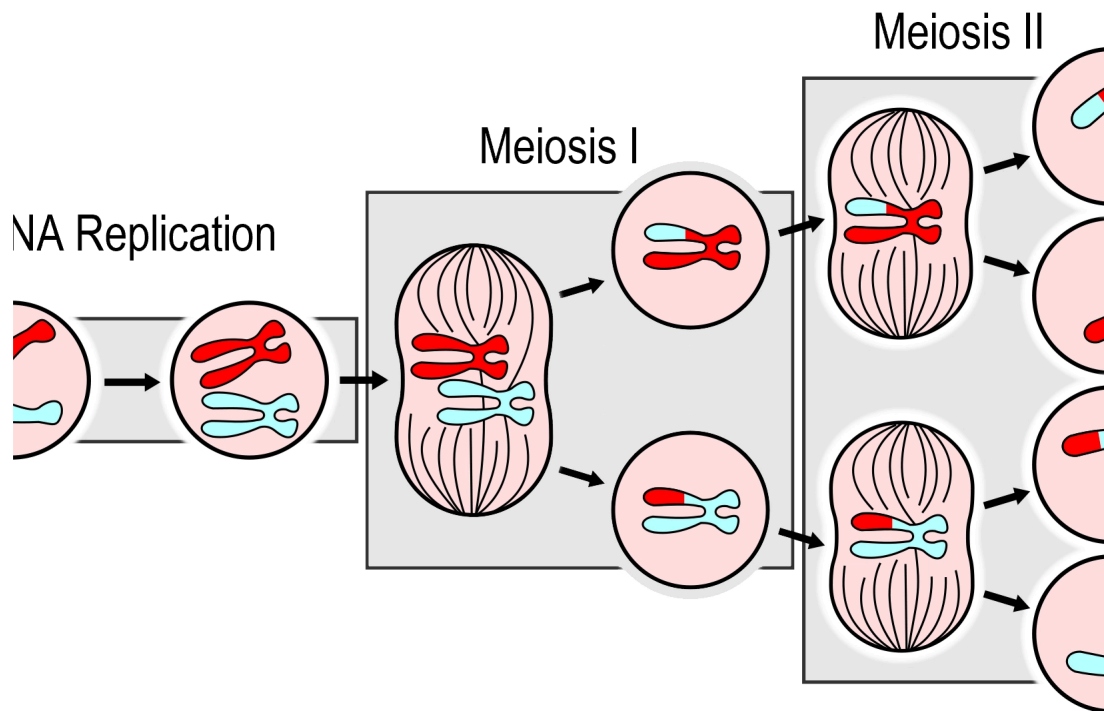
- Each normal adult cell should have 46 chromosomes. In **interphase**, this is **replicated to 92 chromosomes**
- **Homologous chromosomes** (i.e chromosomes of the same number) **line up** and are ready for cell division
 - Note at this stage, **there are four copies of each chromosome** (i.e one paternal + one maternal has become two paternal + two maternal after interphase)
- **Crossing over** occurs between homologous chromosomes from the mother and father which leads to **all chromosomes becoming genetically distinct**
- The **cells split in half**, forming **two daughter cells** each with **46 chromosomes**
 - Each **daughter cell** would have **23 paternal and 23 maternal chromosomes** (though due to crossing over there is now paternal and maternal DNA on each chromosome)

▼ What happens in meiosis II?

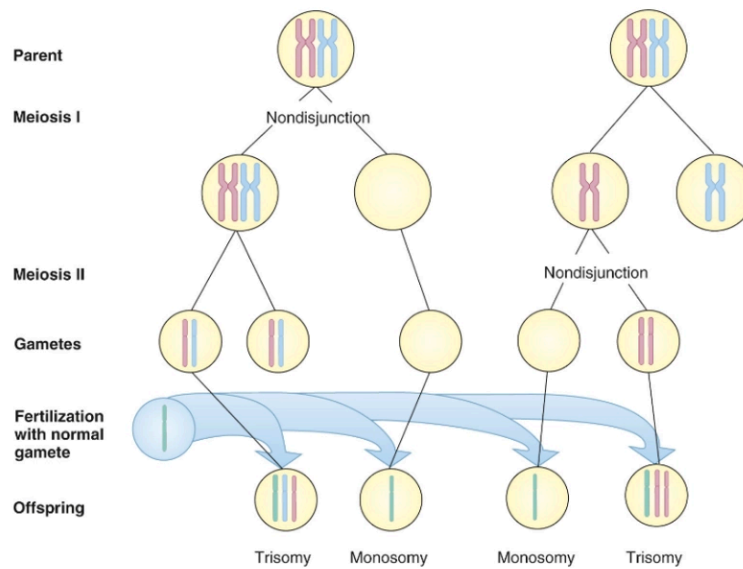
- The cells split in half again now, and **four daughter cells result, each with 23 chromosomes (haploid)**
 - **Each daughter cell would have a mixture of paternal and maternal chromosomes** to allow genetic variation (via random assortment), but **would have 23 chromosomes**, ready to **fuse with another haploid gamete** to form a **diploid zygote**
- Chromosome from meiosis one further splits into two, forming two daughter cells each with one chromatid.

▼ Diagram showing the steps involved in normal meiosis.

- $46 \rightarrow 92 \rightarrow 92 \rightarrow \text{crossing over (46)} \rightarrow 46 \rightarrow 23$



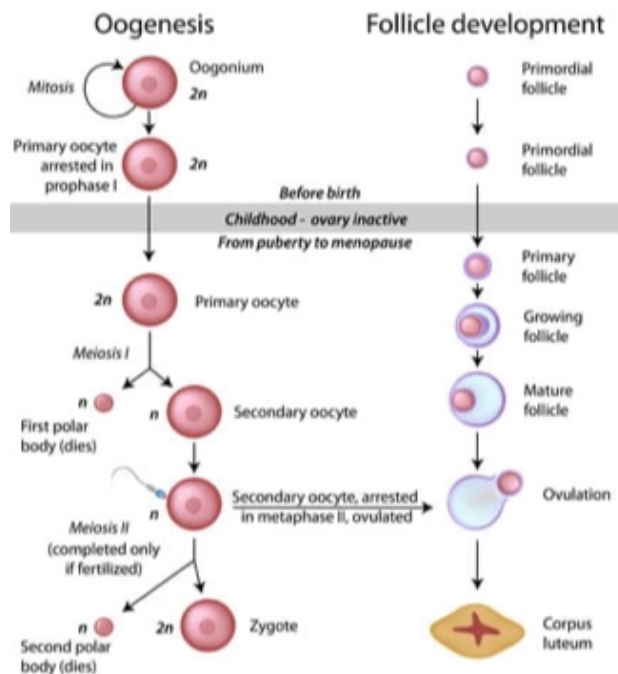
▼ Diagram showing the steps involved in abnormal meiosis.



- ▼ Describe what causes chromosome aneuploidy.
 - This is when the **chromosomes from a parent cell do not split equally** into daughter cells (in meiosis I or II), which results in an **abnormal number of chromosomes within a daughter cell**
 - This process is called **non-disjunction**, when the chromosomes aren't separated in the correct and normal manner.
- ▼ What can X-inactivation of chromosomes lead to?
 - It can lead to only one of the X chromosomes remaining active and the rest (excess) being inactivated
 - It occurs in those with sex chromosome aneuploidy
- ▼ What is a pseudo-autosomal region (PAR)?
 - It is a region that can be present on an X or Y chromosome that can **remain active even after inactivation**
- ▼ Why is it thought that females are more susceptible to autoimmune diseases than males?
 - A lot of the genes from the pseudo-autosomal regions that escape X chromosome inactivation are involved in the immune system, so females have more of these, hence leading to increased susceptibility to autoimmune diseases.
- ▼ What is trisomy-21?
 - A karyotype that involves three chromosomes on the 21st pair instead of two

▼ What maternal factor can influence the likelihood of a baby having trisomy-21?

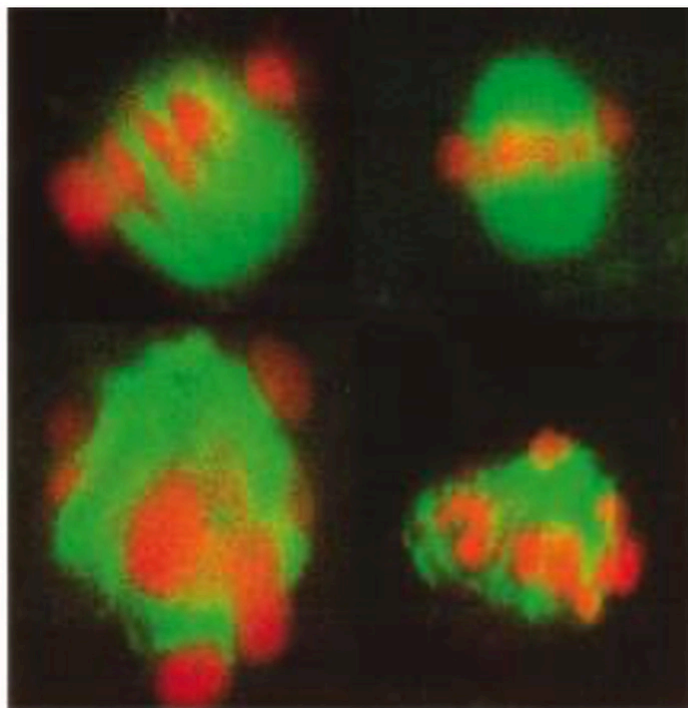
- The **risk of maternal non-disjunction increases with age**, significantly increasing the probability of their offspring developing trisomy-21.



▼ Why is this the case?

- It is likely due to the **degradation of the factors that hold the homologous chromosomes in the middle of the spindle bundle**, that occurs with age.
 - The chromosomes aren't held in the middle of the cell, and this leads to an increased likelihood of cell division in meiosis resulting in an abnormal number of chromosomes in resulting chromosomes
 - Basically, **oogenesis is faulty as women age**.
- Also, the development of a **primary oocyte** → **secondary oocyte** does not occur until a woman's egg is fertilised and it is arrested in prophase
 - This means the egg has more time to accumulate damage, and **abnormalities are increasingly likely**

- ▼ What paternal factor can influence the likelihood of a baby having trisomy-21?
 - **Smoking**
 - Certain **syndromes** such as
 - Apert syndrome
 - Crouzon syndrome
 - Pfeiffer syndrome
- ▼ What is non-disjunction in meiosis?
 - The **failure of the chromosomes to separate**.
- ▼ Image showing how chromosomal non-disjunction occurs increasingly with age.



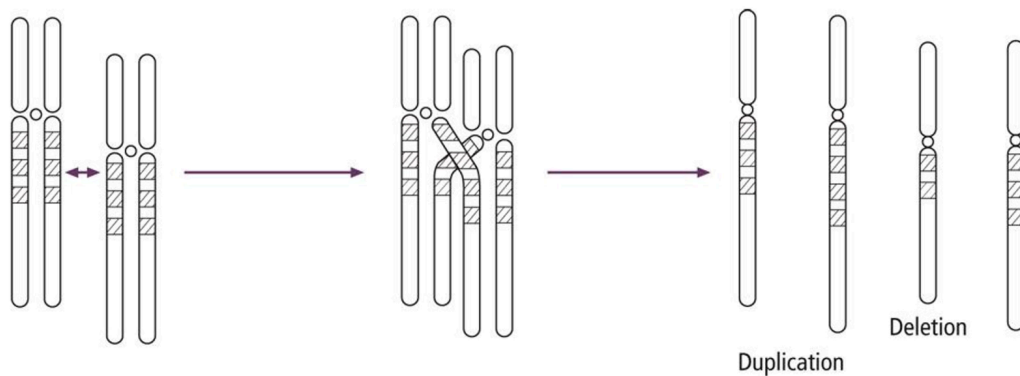
22 year old ♀

40 year old ♀

- ▼ Which is tolerated better, trisomy or monosomy?
 - Trisomy
 - **Most monosomy is incompatible with life**
- ▼ What is mosaicism?
 - The presence of **two or more populations of cells with different genotypes**.

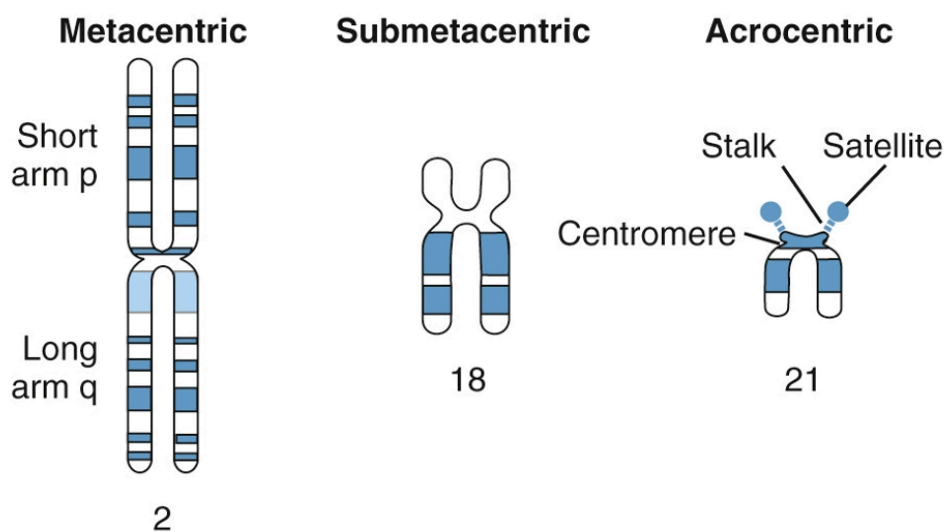
- ▼ Describe what occurs in unequal chromosome crossover and why this is a problem.
- The **chromosomes misalign** when they are lining up to be crossed over, and this causes certain genes to be **replicated** on chromosomes that result from crossing over, and **others to be deleted**

unequal crossover



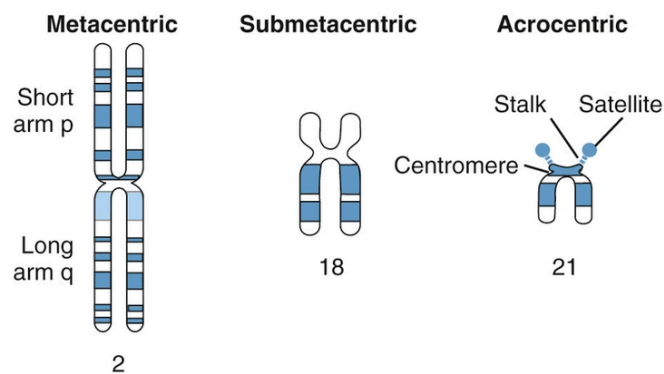
- ▼ What are the three classes of chromosome?

- Metacentric
- Submetacentric
- Acrocentric



▼ What is Robertsonian translocation?

- Robertsonian translocations are **a specific class of translocations in which two acrocentric chromosomes fuse at their centric ends**
- It can occur on chromosomes **13,14,15 and 21** and **can cause problems in offspring.**
- It involves **part or the entirety of these chromosomes being moved onto another chromosome**



▼ ROBERTSONIAN TRANSLOCATION IN REPRODUCTION.

- **NOTE:** In this image, the "segregation products" refer to the chromosomes that will be seen in the sperm or egg haploid cell
- If a person has Robertsonian translocation of chromosome 14 and 21, this will form a derivative of the two as seen below